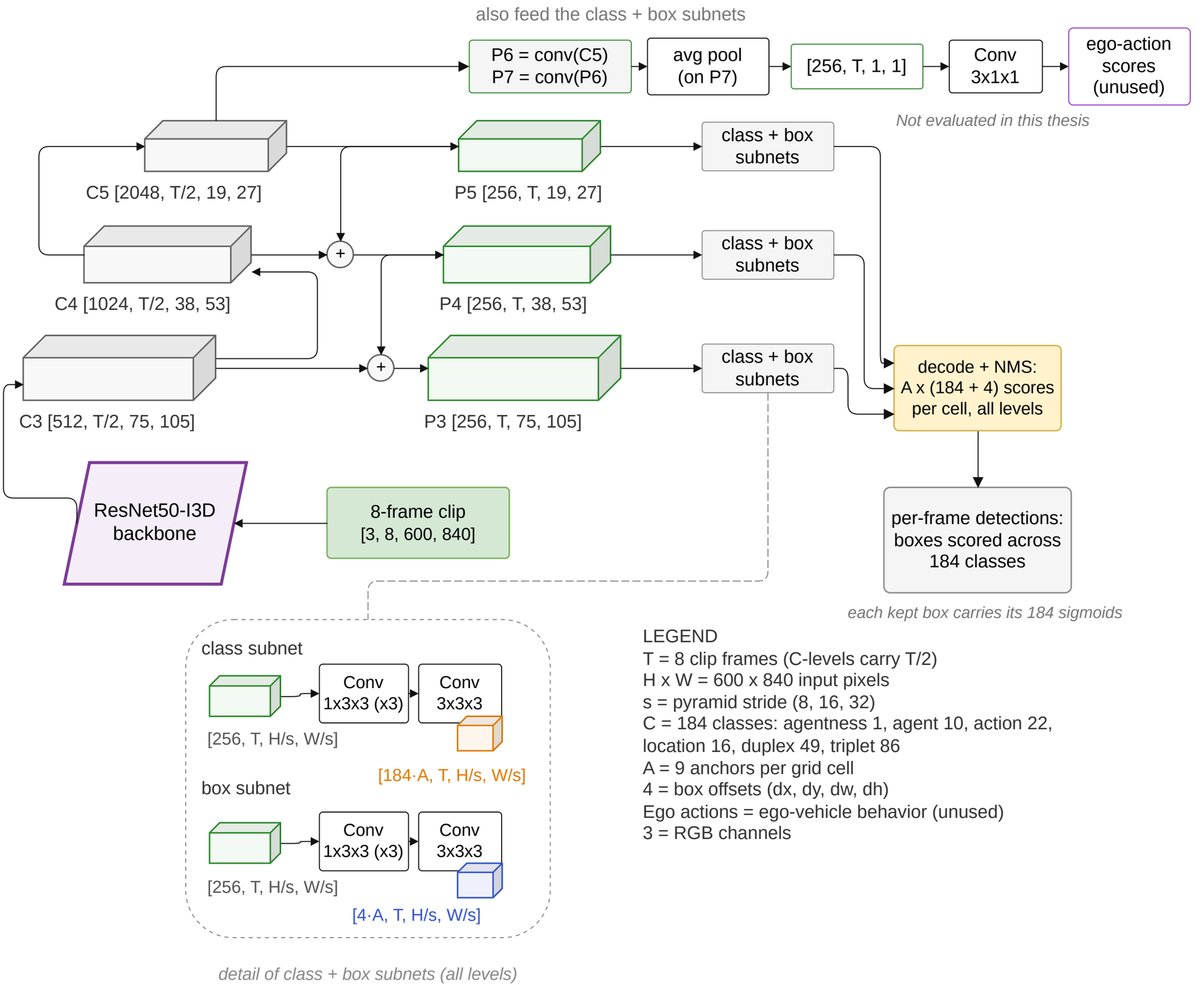

also feed the class + box subnets
P6 = conv(C5)
P7 = conv(P6)
avg pool
(on P7)
[256, T, 1, 1]
Conv
3x1x1
ego-action
scores
(unused)
Not evaluated in this thesis
C5 [2048, T/2, 19, 27]
P5 [256, T, 19, 27]
class + box
subnets
C4 [1024, T/2, 38, 53]
P4 [256, T, 38, 53]
class + box
subnets
C3 [512, T/2, 75, 105]
P3 [256, T, 75, 105]
class + box
subnets
decode + NMS:
A x (184 + 4) scores
per cell, all levels
ResNet50-I3D
backbone
8-frame clip
[3, 8, 600, 840]
per-frame detections:
boxes scored across
184 classes
each kept box carries its 184 sigmoids
class subnet
Conv
1x3x3 (x3)
Conv
3x3x3
[256, T, H/s, W/s]
[184·A, T, H/s, W/s]
box subnet
Conv
1x3x3 (x3)
Conv
3x3x3
[256, T, H/s, W/s]
[4·A, T, H/s, W/s]
detail of class + box subnets (all levels)
LEGEND
T = 8 clip frames (C-levels carry T/2)
H x W = 600 x 840 input pixels
s = pyramid stride (8, 16, 32)
C = 184 classes: agentness 1, agent 10, action 22,
location 16, duplex 49, triplet 86
A = 9 anchors per grid cell
4 = box offsets (dx, dy, dw, dh)
Ego actions = ego-vehicle behavior (unused)
3 = RGB channels